Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT I : Descriptive Statistics

Syllabus: Statistics introduction, population vs sample, collection of data (primary, secondary), measures of central tendency, measures of variability, skewness, kurtosis, correlation, correlation coefficient, rank correlation, regression coefficients, method of least squares, regression lines.

I. Introduction to Statistics

Definition:

Statistics is the science of collecting, organising, presenting, analysing and interpreting numerical data to help in decision making.

- The word comes from the Latin ***status*** / Italian ***statista*** / German ***Statistik*** = "political state".
- Father of Statistics: ***R. A. Fisher***; early work by Sir Francis Galton and Karl Pearson.
- Plural sense: numerical facts (data). Singular sense: the science / methods.

Characteristics of statistics

- Aggregate of facts (a single number is not statistics).
- Numerically expressed and affected by many causes.
- Collected in a systematic manner for a pre-determined purpose.
- Comparable with other data of the same kind.

Functions of statistics

- Simplifies complexity - condenses large data into averages, tables, graphs.
- Presents facts in a definite form; helps in comparison.
- Helps in forecasting, planning and policy making.
- Helps in testing hypotheses and formulating theories.

Limitations

- Deals with groups, not individuals.
- Only quantitative data; results are true on average only.
- Can be misused (wrong sampling, wrong conclusions).

Types of Statistics

Descriptive	Inferential
Summarises data actually collected	Draws conclusions about population from
Mean, median, SD, graphs, correlation	Estimation, hypothesis tests, regression
No generalisation	prediction Generalises with probability

Note:

This course: Unit I = descriptive; Units II-III = probability theory; Units IV-V = inference.

2. Population vs Sample

Definition:

Population: the complete collection of all items / individuals under study.

Sample: a part (subset) of the population selected to represent it.

- Finite population: countable (students in a college). Infinite: uncountable (all possible coin tosses).
- **Parameter**: a numerical measure of a population (μ, σ, P).
- **Statistic**: a numerical measure of a sample ($\bar{x}$, s, p).

Basis	Population	Sample
Size	N (all units)	n (part of N)
Measure called	Parameter	Statistic
Symbols	μ, σ^2, P	$\bar{x}$, s^2, p

Value	Fixed (usually unknown)	Varies sample to sample
Study type	Census / complete enumeration	Sample survey

Census vs Sampling

- Census: every unit is studied - accurate but costly and slow.
- Sampling: faster, cheaper, needed when testing is destructive (bulbs, crash tests).

Methods of sampling (brief)

- ***Simple random***: every unit has equal chance (lottery / random numbers).
- ***Stratified***: divide into homogeneous strata, sample from each.
- ***Systematic***: pick every k-th unit, k = N/n.
- ***Cluster***: choose whole groups (clusters) randomly.
- Non-random: judgement, quota, convenience sampling.

> **Note:**
> Sample must be random and representative, otherwise it gives biased results.

3. Collection of Data

- Data = raw facts / figures. Two sources: primary and secondary.

Primary data

> **Definition:**
> Data collected first-hand by the investigator for the first time, for a specific purpose. It is original.

Methods of collecting primary data

- Direct personal investigation (interview).
- Indirect oral investigation (asking witnesses / third persons).

- Questionnaires: mailed, or filled by enumerators / schedules.
- Local sources / correspondents (newspapers, agencies).
- Observation and experiments.

Features of a good questionnaire

- Short, simple, clear and unambiguous questions.
- Logical order; avoids personal / leading questions.
- Has instructions and a covering letter; pre-tested by a pilot survey.

Secondary data

Definition:
Data already collected by someone else, and available in published or unpublished form.

- Published: government reports (Census, NSSO, RBI bulletin), journals, newspapers, research papers, UN / World Bank reports.
- Unpublished: records of companies, universities, research workers.
- Must be checked for suitability, reliability and adequacy before use.

Primary data	Secondary data
Original, first-hand	Already available, second-hand
More accurate for purpose	May not suit the purpose
Costly and time consuming	Cheap and quick
Needs large manpower	Little manpower
Example: own survey	Example: census report

Classification and presentation

- Raw data is arranged as frequency distribution: discrete (values) or continuous (class intervals).
- Class width h, class mark = (lower + upper) / 2, cumulative frequency (less than / more than).

- Presented by tables, histogram, frequency polygon, ogive, pie and bar diagrams.

4. Measures of Central Tendency

Definition:

An average is a single value that represents the whole data and lies near its centre. A good average is rigidly defined, simple, based on all items, and not affected by extremes.

- Types: Arithmetic mean (AM), Median, Mode, Geometric mean (GM), Harmonic mean (HM).

(a) Arithmetic Mean

Formula:

Ungrouped: $\bar{X} = \Sigma x / n$

Discrete / grouped: $\bar{X} = \Sigma fx / \Sigma f$ (x = value or class mark)

Step deviation: $\bar{X} = A + h \cdot (\Sigma fu / \Sigma f)$, $u = (x - A) / h$

- Properties: $\Sigma(x - \bar{X}) = 0$; $\Sigma(x - \bar{X})^2$ is minimum; affected by extreme values.
- Combined mean of two groups: $\bar{X} = (n1 \cdot \bar{X}1 + n2 \cdot \bar{X}2) / (n1 + n2)$.
- Weighted mean: $\bar{X}w = \Sigma wx / \Sigma w$.

(b) Median (middle value)

- Ungrouped: arrange in order; n odd → value of ((n+1)/2)-th item; n even → average of two middle items.

Formula:

Grouped: Median = $L + [(N/2 - cf) / f] \times h$

- L = lower limit of median class, cf = cumulative frequency before it, f = its frequency, h = width.
- Not affected by extremes; found graphically by ogive. Quartiles Q1, Q3 and deciles use N/4, 3N/4, ...

(c) Mode (most frequent value)

Formula:

Mode = $L + [(f1 - f0) / (2f1 - f0 - f2)] \times h$

- f1 = frequency of modal class, f0 = before it, f2 = after it.
- A distribution may be unimodal, bimodal or multimodal.

Formula:

Empirical relation: Mode = 3 Median - 2 Mean (moderately skewed data)

(d) Geometric and Harmonic means

Formula:

$GM = (x1 \cdot x2 \cdot \ldots \cdot xn)^{1/n}$; $\log GM = \Sigma \log x / n$

Formula:

$HM = n / \Sigma(1/x)$ [grouped: $\Sigma f / \Sigma(f/x)$]

Note:

For positive unequal values: AM > GM > HM, and $GM^2 = AM \times HM$ (for two values).

- GM used for growth rates, ratios, index numbers; HM used for average speed, rates.

Example:

Find mean, median, mode for the data.

Class: 0-10, 10-20, 20-30, 30-40, 40-50 ; f = 5, 8, 15, 12, 10 (N = 50)

Class	f	x (mark)	fx	cf
0-10	5	5	25	5
10-20	8	15	120	13
20-30	15	25	375	28
30-40	12	35	420	40
40-50	10	45	450	50

- $\Sigma fx = 1390 \rightarrow$ Mean = 1390 / 50 = **27.8**
- N/2 = 25 → median class 20-30 : Median = $20 + (25 - 13)/15 \times 10 = 20 + 8$ = **28**
- Modal class 20-30 : Mode = $20 + (15 - 8)/(30 - 8 - 12) \times 10 = 20 + 7$ = **27**

Note:

Which average to use? Mean for symmetric data, median for skewed/open-ended, mode for most typical value.

5. Measures of Variability (Dispersion)

Definition:

Dispersion measures how much the observations scatter around the average. Two sets can have equal mean but different spread.

- Absolute measures have units; relative measures (coefficients) are pure numbers used to compare.

Types

Measure	Formula	Relative measure
Range	L - S	(L - S) / (L + S)
Quartile deviation	(Q3 - Q1) / 2	(Q3 - Q1) / (Q3 + Q1)
Mean deviation	$\Sigma\|x - A\| / n$	MD / A
Standard deviation	σ	CV = $(\sigma / \bar{x}) \times 100$

Range

Formula:

Range = Largest - Smallest value

- Simplest; depends only on two extreme values.

Mean deviation

Formula:

MD about A = $\Sigma f|x - A| / \Sigma f$ (A = mean or median)

- Least when taken about the median.

Variance and Standard Deviation

Formula:

$\sigma^2 = \Sigma(x - \bar{x})^2 / n = \Sigma x^2/n - (\bar{x})^2$

Grouped: $\sigma^2 = \Sigma f(x - \bar{x})^2 / \Sigma f = \Sigma fx^2/\Sigma f - (\bar{x})^2$

$\sigma = \sqrt{\text{variance}}$; sample variance $s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$

- Properties: independent of change of origin; depends on change of scale: $\sigma(a + bx) = |b|\sigma$.
- Combined SD: $\sigma^2 = [n1(\sigma1^2 + d1^2) + n2(\sigma2^2 + d2^2)] / (n1 + n2)$, $d = \bar{x}i - \bar{x}$.
- Coefficient of variation CV = $(\sigma / \bar{x}) \times 100$ %. Lower CV → more consistent (stable) series.

Example:

Data: 2, 4, 4, 4, 5, 5, 7, 9. Find mean, MD, variance, SD.

Mean = 40/8 = 5. $\Sigma|x - 5| = 3+1+1+1+0+0+2+4 = 12 \rightarrow MD = 12/8 = 1.5$

$\Sigma(x - 5)^2 = 9+1+1+1+0+0+4+16 = 32 \rightarrow \sigma^2 = 4$, $\sigma = 2$, $CV = 2/5 \times 100 = 40\ \%$

Example:

Continue the grouped data of the earlier example (mean = 27.8).

$\Sigma f(x - 27.8)^2 = 7608 \rightarrow \sigma^2 = 7608/50 = 152.16$, $\sigma = 12.34$, $CV = 44.4\ \%$

6. Moments, Skewness and Kurtosis

Moments

Formula:

r-th central moment: $\mu r = \Sigma f\,(x - \bar{x})^r / \Sigma f$

- $\mu 0 = 1$, $\mu 1 = 0$, $\mu 2 = \sigma^2$ (variance).

Formula:

Raw moments about A: $\mu r' = \Sigma f\,(x - A)^r / \Sigma f$

Formula:

$\mu 2 = \mu 2' - (\mu 1')^2$; $\mu 3 = \mu 3' - 3\mu 2'\mu 1' + 2(\mu 1')^3$

$\mu 4 = \mu 4' - 4\mu 3'\mu 1' + 6\mu 2'(\mu 1')^2 - 3(\mu 1')^4$

Skewness

Definition:

Skewness is the lack of symmetry of a distribution.

- Symmetric: mean = median = mode (bell shaped).
- **Positive** skew (tail to right): mean > median > mode.
- **Negative** skew (tail to left): mean < median < mode.

Type	Relation	Sk
Symmetric	$\bar{X} = M = Z$	0
Positively skewed	$\bar{X} > M > Z$	positive
Negatively skewed	$\bar{X} < M < Z$	negative

Measures of skewness

Formula:

Karl Pearson: Sk = (Mean - Mode) / σ or 3(Mean - Median) / σ (limits ±3)

Formula:

Bowley (quartile): Sk = (Q3 + Q1 - 2·Median) / (Q3 - Q1) (limits ±1)

Formula:

Moment measure: $\beta_1 = \mu_3^2 / \mu_2^3$; $\gamma_1 = \sqrt{\beta_1} = \mu_3 / \sigma^3$ (sign of μ_3)

Example:

For the grouped data above: Sk = (27.8 - 27) / 12.34 = 0.065 → almost symmetric, slight positive skew.

Kurtosis

Definition:

Kurtosis measures the peakedness (flatness) of a frequency curve compared with the normal curve.

Formula:

$\beta_2 = \mu_4 / \mu_2^2$; $\gamma_2 = \beta_2 - 3$

Type	β_2	Shape
Mesokurtic	= 3	Normal curve
Leptokurtic	> 3	More peaked, thin tails
Platykurtic	< 3	Flat topped

Note:

For the normal distribution: $\mu_3 = 0$, $\beta_1 = 0$ and $\beta_2 = 3$.

7. Correlation

Definition:

Correlation is the statistical relationship between two variables such that a change in one is accompanied by a change in the other.

- **Positive**: both move in same direction (height - weight).
- **Negative**: move in opposite directions (price - demand).
- Linear / non-linear; simple / multiple / partial.
- Methods: scatter diagram, Karl Pearson's coefficient, Spearman's rank correlation, concurrent deviation.

Scatter diagram

- Points rising upward → positive; falling → negative; no pattern → zero correlation; points on a line → perfect correlation ($r = \pm 1$).

Karl Pearson's coefficient of correlation

Formula:

$r = Cov(x, y) / (\sigma x \cdot \sigma y)$

$Cov(x, y) = \Sigma(x - \bar{x})(y - \bar{y})/n = \Sigma xy/n - \bar{x}\bar{y}$

$r = [n\Sigma xy - (\Sigma x)(\Sigma y)] / \sqrt{\{[n\Sigma x^2 - (\Sigma x)^2] [n\Sigma y^2 - (\Sigma y)^2]\}}$

Properties of r

- $-1 \leq r \leq 1$; r is a pure number (no unit).
- Independent of change of origin and scale (same sign for positive scale).
- Symmetric: $r_{xy} = r_{yx}$.
- r = 0 means no linear relation (not necessarily independence). Independent variables $\Rightarrow$ r = 0.
- Coefficient of determination = r^2; probable error P.E. = $0.6745 (1 - r^2)/\sqrt{n}$.

r value	Interpretation
+1 / -1	Perfect positive / negative
0.7 to 0.99	High degree
0.3 to 0.7	Moderate
0 to 0.3	Low / none

Example:

Find r for x: 1, 2, 3, 4, 5 y: 2, 4, 5, 4, 5

$n = 5, \Sigma x = 15, \Sigma y = 20, \Sigma xy = 66, \Sigma x^2 = 55, \Sigma y^2 = 86$

$r = (5 \times 66 - 15 \times 20) / \sqrt{[(275 - 225)(430 - 400)]} = 30 / \sqrt{1500} = \mathbf{0.775}$ (high positive)

8. Spearman's Rank Correlation

Definition:

Used when data is given as ranks, or when variables are qualitative (beauty, honesty, ability).

Formula:

$\rho = 1 - 6\Sigma d^2 / [n(n^2 - 1)]$, $d = R1 - R2$

- $-1 \leq \rho \leq 1$. $\rho = 1$ identical ranking; $\rho = -1$ exactly opposite ranking; $\Sigma d = 0$ (check).

Repeated (tied) ranks

- Give each tied item the average of the ranks it occupies (two items at rank 4, 5 each get 4.5).

Formula:

$\rho = 1 - 6\,[\Sigma d^2 + \Sigma(m^3 - m)/12] / [n(n^2 - 1)]$

- m = number of times a value repeats; add one term per repeated group.

Example:

Marks in two tests (9 students): find ρ.

X	35	23	47	17	10	43	9	6	28
Y	30	33	45	23	8	49	12	4	31
Rank	3	5	1	6	7	2	8	9	4
Rank Y	5	3	2	6	8	1	7	9	4
d	-2	2	-1	0	-1	1	1	0	0
d^2	4	4	1	0	1	1	1	0	0

- $\Sigma d^2 = 12$, $n = 9 \rightarrow \rho = 1 - (6 \times 12)/(9 \times 80) = 1 - 0.1 =$ **0.9** (strong agreement)

Note:

Pearson's r uses actual values; Spearman's ρ uses ranks. For no ties ρ equals Pearson's r computed on the ranks.

9. Regression

Definition:

Regression is the statistical method of estimating (predicting) the value of one variable from the known value of another using their average relationship.

- The line of regression of y on x predicts y; of x on y predicts x. Term given by Francis Galton.

Correlation	Regression
Measures degree of relation	Measures nature / form (prediction)
r symmetric in x, y	Regression of y on x ≠ x on y
No cause-effect needed	Dependent and independent variable

Method of least squares

- Fit $y = a + bx$ so that $\Sigma e^2 = \Sigma(y - a - bx)^2$ is minimum.
- Setting partial derivatives to zero gives the ***normal equations***:

Formula:

$$\Sigma y = na + b\Sigma x$$
$$\Sigma xy = a\Sigma x + b\Sigma x^2$$

- Solve for a and b. (Similarly for x on y: $\Sigma x = na + b\Sigma y$, $\Sigma xy = a\Sigma y + b\Sigma y^2$.)

Regression lines and coefficients

Formula:

y on x: $y - \bar{y} = b_{yx}(x - \bar{x})$, $b_{yx} = r\,\sigma y / \sigma x = \mathrm{Cov}(x, y) / \sigma x^2$

Formula:

x on y: $x - \bar{x} = b_{xy}(y - \bar{y})$, $b_{xy} = r\,\sigma x / \sigma y = \mathrm{Cov}(x, y) / \sigma y^2$

Short form: $b_{yx} = [n\Sigma xy - \Sigma x \Sigma y] / [n\Sigma x^2 - (\Sigma x)^2]$

Properties

- Both lines pass through the point $(\bar{x}, \bar{y})$ - their intersection gives the means.
- $r^2 = b_{yx} \cdot b_{xy}$ so $r = \pm\sqrt{(b_{yx}\, b_{xy})}$; r, b_{yx}, b_{xy} have the same sign.
- If one coefficient > 1, the other must be < 1 (product ≤ 1).
- AM of regression coefficients ≥ r.
- Regression coefficients are independent of change of origin but not of scale.
- If $r = \pm 1$ the two lines coincide; if $r = 0$ they are perpendicular ($y = \bar{y}$ and $x = \bar{x}$).

Formula:

Angle between lines: $\tan\theta = [(1 - r^2) / r] \times [\sigma x \sigma y / (\sigma x^2 + \sigma y^2)]$

Example:

For x: 1, 2, 3, 4, 5 and y: 2, 4, 5, 4, 5 find both regression lines.

$\bar{x} = 3$, $\bar{y} = 4$, Cov = 66/5 - 12 = 1.2, $\sigma x^2 = 2$, $\sigma y^2 = 1.2$

$b_{yx} = 1.2/2 = 0.6 \rightarrow y - 4 = 0.6(x - 3) \rightarrow \mathbf{y = 0.6x + 2.2}$

$b_{xy} = 1.2/1.2 = 1 \rightarrow x - 3 = 1(y - 4) \rightarrow \mathbf{x = y - 1}$

Check: $r^2 = 0.6 \times 1 = 0.6 \rightarrow r = 0.775$ ✓. Prediction: $x = 6 \rightarrow y = 5.8$.

Note:

Exam tips: always check ranks sum to $n(n+1)/2$, remember $r = \sqrt{(b_{yx}\, b_{xy})}$ with common sign, and write the formula first.

Quick Revision - Unit I

- Mean = $\Sigma fx/\Sigma f$, Median = $L + (N/2 - cf)h/f$, Mode = $L + (f1-f0)h/(2f1-f0-f2)$.
- $\sigma^2 = \Sigma fx^2/\Sigma f - \bar{x}^2$; $CV = \sigma/\bar{x} \times 100$.
- $Sk = (Mean - Mode)/\sigma$; $\beta 2 = \mu 4/\mu 2^2$ (3 = normal).
- $r = Cov/(\sigma x \sigma y)$; $\rho = 1 - 6\Sigma d^2/n(n^2 - 1)$.
- Regression lines meet at $(\bar{x}, \bar{y})$; $r^2 = b_{yx}\, b_{xy}$.